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# Research on Intelligent Crop Pest and Disease Recognition Methods Integrating Multi-Layer Attention Mechanisms

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***Abstract*—Crop pests and diseases are critical factors affecting agricultural production and food security, making early and accurate monitoring essential for effective control. To address the issues of low efficiency and limited detection capabilities in traditional manual monitoring, this paper proposes a YOLOv8 crop pest and disease detection model that integrates the SimAM attention mechanism. This approach embeds SimAM modules at layers 9, 17, and 21 of the YOLOv8 network, achieving multi-scale adaptive enhancement of shallow-level details, mid-level semantics, and deep-level features to improve the model's perception of critical pest and disease areas. Validation on a grape leaf disease dataset demonstrates that the improved model achieves a recall of 0.842 (up from 0.810), mAP@0.5of 0.823 (up from 0.816), and mAP@0.5:0.95of 0.534 (up from 0.528) compared to the original YOLOv8. Experimental results demonstrate that the YOLOv8-SimAM model enhances pest and disease detection accuracy and robustness while maintaining lightweight performance, providing an effective technical solution for intelligent early monitoring and precise control of crop pests and diseases.**

***Keywords*—*Crop Pests and Diseases, Object Detection, Attention Mechanism***

## I. INTRODUCTION

Crop pests and diseases are one of the key factors constraining agricultural production and food security. Their occurrence is characterized by strong concealment, rapid transmission, and significant regional variations. Failure to accurately identify and effectively control these pests and diseases in their early stages often leads to reduced yields or even total crop failure, causing severe losses to the agricultural economy and ecological environment. In recent years, the rapid advancement of computer vision and deep learning technologies has opened new technical avenues for intelligent identification of crop pests and diseases. Object detection algorithms based on convolutional neural networks can automatically learn multi-level features from images, enabling precise localization and recognition of targets such as disease lesions and insect bodies in complex backgrounds[1]. Archishman et al. [2] developed a crop leaf pest and disease recognition model centered on ResNet50 using deep learning and machine learning techniques, achieving high-precision identification of pest and disease types.

Among numerous object detection algorithms, the YOLO series has garnered significant attention for its end-to-end architecture and outstanding real-time performance. As the latest generation of single-stage object detection models, YOLOv8 has been optimized in aspects such as network architecture design, feature fusion methods, and training strategies. It further enhances inference speed while maintaining detection accuracy, demonstrating strong potential for engineering deployment. Liangquan Jia et al.[3] proposed a lightweight rice pest and disease detection model based on an improved YOLOv7. They introduced MobileNetV3 as the feature extraction network to reduce model complexity and integrated the Coordinate Attention mechanism with the SIoU loss function to enhance feature representation and localization accuracy. Jiao Wang et al.[4] addressed computational intensity and resource constraints in mobile deployment of mango pest detection models by proposing the lightweight object detection model GAS-YOLOv8. By replacing the backbone network with GhostHGNetv2, introducing the lightweight detection head AsDDet, and substituting the C2f module with C2f-SE, they significantly reduced model parameters while improving detection accuracy. Hao Sun et al.[5] addressed limitations in edge device deployment for vegetable pest and disease detection by proposing YOLO-FMDI, a lightweight, high-precision object detection model based on YOLOv8n. By designing a multi-scale feature diffusion interaction neck structure, an efficient multi-core feature extraction network, and introducing a universal inverse residual bottleneck module, the model significantly reduces complexity while maintaining detection accuracy.

Attention mechanisms, a prominent research focus in recent deep learning studies, adaptively weight features to guide the network toward more discriminative regions and channel information, thereby effectively enhancing the model's feature representation capabilities[6]. SimAM is a parameter-free attention mechanism that measures neuron importance based on an energy function. Requiring no additional trainable parameters, it offers advantages such as simple structure, low computational overhead, and ease of integration, making it particularly suitable for lightweight and real-time-demanding object detection models.

Building upon this, this paper focuses on the task of intelligent crop pest and disease detection, exploring deep learning-based detection methods. We propose a YOLOv8-based crop pest and disease detection model that integrates the SimAM attention mechanism. By incorporating the SimAM module into the YOLOv8 architecture, the model's ability to focus on critical pest/disease regions and fine-grained features is enhanced, thereby improving detection accuracy and robustness in complex agricultural scenarios. This research aims to provide an efficient and reliable technical solution for the early monitoring and precise control of crop pests and diseases. It holds theoretical significance and practical application value for advancing smart agriculture and agricultural informatization.

## II. Related Theories

### A. Object Detection Algorithm

Object detection algorithms represent one of the core technologies in computer vision, aiming to simultaneously identify and localize specific objects within images or videos. This task requires addressing two key challenges: determining whether an object exists in an image and pinpointing its precise location within the image[7]. Traditional approaches primarily relied on manually designed features and classifiers. With the advancement of deep learning, modern object detection algorithms have evolved into two main paradigms: two-stage detection algorithms, which first generate candidate regions before performing classification and regression; and single-stage detection algorithms, which treat detection as an end-to-end regression problem, achieving a better balance between speed and accuracy. Furthermore, Transformer-based detectors leverage self-attention mechanisms for global modeling, further enhancing detection performance. Object detection algorithms find extensive applications in real-world scenarios such as autonomous driving, video surveillance, and medical image analysis. Core evaluation metrics include accuracy and efficiency, with ongoing research focused on achieving more robust and efficient detection under complex conditions like occlusion and scale variations.

### B. YOLOv8 Algorithm

YOLOv8 is one of the latest representatives in the YOLO series of object detection algorithms. Its core philosophy inherits the fundamental concept of the YOLO family, treating object detection as a single, end-to-end regression problem. It achieves extremely high detection speeds by directly predicting object bounding boxes, class probabilities, and confidence scores across the entire image during a single forward pass[8].

Compared to its predecessors, YOLOv8's core improvements and advantages are primarily reflected in the following aspects:

- It adopts more efficient backbone networks like CSPDarknet or similar designs, combined with the advanced PAN-FPN structure for multi-scale feature fusion. This significantly enhances detection capabilities for objects of varying scales, demonstrating particularly outstanding performance in small object detection.
- YOLOv8 abandons predefined anchor boxes, instead employing a decoupled head to directly predict the offset of bounding boxes relative to the grid center. Its core bounding box regression formula can be simplified as:

$$b_x = \sigma(t_x) + c_x \tag{1}$$

$$b_y = \sigma(t_y) + c_y \tag{2}$$

$$b_w = p_w \cdot e^{t_w} \tag{3}$$

$$b_h = p_h \cdot e^{t_h} \tag{4}$$

  $(c_x, c_y)$ denotes the coordinates of the upper-left corner of the grid, $(t_x, t_y, t_w, t_h)$ represents the model's predicted value, $\sigma$ is the Sigmoid function, and $(p_w, p_h)$ is the scale factor propagated from the previous layer. This design simplifies the training process and reduces reliance on prior knowledge.

- The loss function of YOLOv8 typically consists of classification loss, bounding box regression loss, and object confidence loss. Among these, bounding box regression commonly employs CIoU Loss or its variants, which comprehensively consider the area of intersection, center point distance, and aspect ratio:

$$\mathcal{L}_{CIoU} = 1 - IoU + \frac{\rho^2(b, b^{gt})}{c^2} + \alpha v \tag{5}$$

  Here, $IoU$ represents the intersection-over-union ratio, $\rho$ denotes the Euclidean distance to the center point, $c$ is the diagonal length of the minimum bounding rectangle, and $v$ serves as a parameter measuring aspect ratio consistency. This loss function enhances the precision of bounding box localization.

## III. Methodology

### A. SimAM Attention Mechanism

Attention mechanisms enhance a model's representational capacity and detection performance by guiding the network to focus on task-critical regions or feature channels through adaptive weighting of features. Existing mainstream attention mechanisms typically rely on additional trainable parameters or convolutional operations. While improving performance, they inevitably increase model complexity and computational overhead, hindering lightweight model development and edge device deployment.

SimAM is a parameter-free attention mechanism. Its core idea is to model the importance of neurons through an energy function, achieving adaptive enhancement of key neurons in feature maps without introducing any additional learnable parameters. This approach balances performance improvement with computational efficiency[9].

SimAM treats each neuron in the feature map as an independent information unit, evaluating its importance by measuring its separability from other neurons. For an input feature map $\mathbf{X} \in \mathbb{R}^{C \times H \times W}$, where $C$, $H$, and $W$ denote the number of channels, height, and width, respectively, the energy function for a neuron $x_i$ in the $c$ channel is defined as:

$$E(x_i) = \frac{(x_i - \mu)^2}{\sigma^2 + \varepsilon} \tag{6}$$

Where $\mu$ and $\sigma^2$ denote the mean and variance of all neurons within the channel, respectively, and $\varepsilon$ is a minimal constant preventing the denominator from becoming zero. This energy function reflects the degree of deviation of the neuron from the overall distribution of the channel. Greater deviation indicates higher distinguishability and greater importance for the target task.

Figure 1 lists three existing types of attention modules. Channel attention is 1D attention, treating different channels differentially while treating all positions equally. Spatial attention is 2D attention, treating different positions differentially while treating all channels equally. SimAM is 3D attention, inferring 3D attention weights for feature maps within a layer without adding parameters to the original network. SimAM can learn long-range dependencies in input sequences and generate corresponding output sequences.

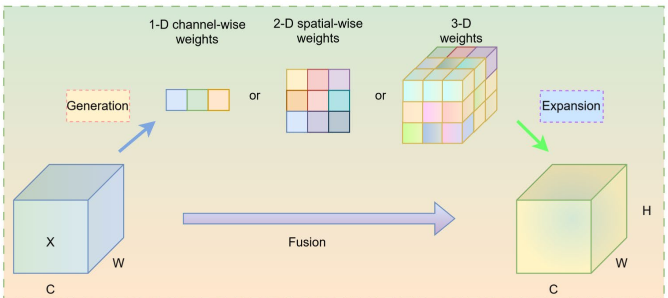


Fig. 1. Comparisons of different attention steps

## B. Integration Approach

YOLOv8 adopts a hierarchical structure comprising Backbone–Neck–Head layers, with distinct differences in semantic information and spatial resolution across each level. Shallow features contain rich spatial detail, mid-level features combine semantic and positional information, while deep features exhibit strong semantic representation capabilities.

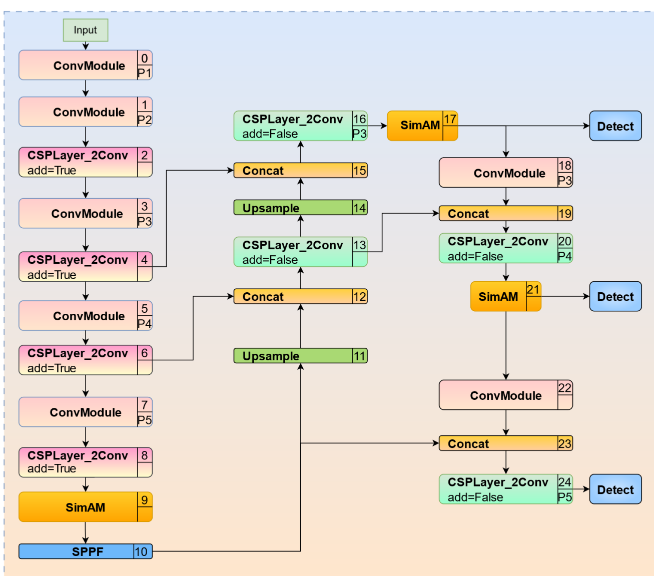


Fig. 2. YOLO Architecture Incorporating SimAM Attention Mechanism

This paper employs a multi-layer embedded SimAM attention mechanism, introducing SimAM modules at layers 9, 17, and 21 of the YOLOv8 network to adaptively enhance features at different levels. This enables multi-scale modeling of fine-grained pest and disease features. By integrating SimAM modules at layers 9, 17, and 21, the model establishes a synergistic enhancement mechanism across shallow, intermediate, and deep feature spaces, enabling comprehensive attention to pest targets from local details to high-level semantics. Since SimAM introduces no additional learnable parameters, this multi-layer embedding strategy effectively enhances YOLOv8's feature representation capability and detection performance for crop pest and disease detection tasks with minimal increase in model complexity and computational overhead. Figure 2 shows the overall network architecture, where the orange boxes denote the integrated SimAM attention modules.

# IV. Experiments and Analysis

## A. Experimental Dataset and Evaluation Metrics

The agricultural dataset used in this paper is the grape leaf disease dataset, annotated in YOLO format into three categories: black rot, black measles, and blight. The dataset comprises 2,306 images, divided into a training set and a test set in a 7:3 ratio. The evaluation metrics employed in this study primarily include Precision, Recall, F1-Score, mAP@0.5, and mAP@0.5:0.95[10]. The target distribution of the dataset is shown in Figure 3.

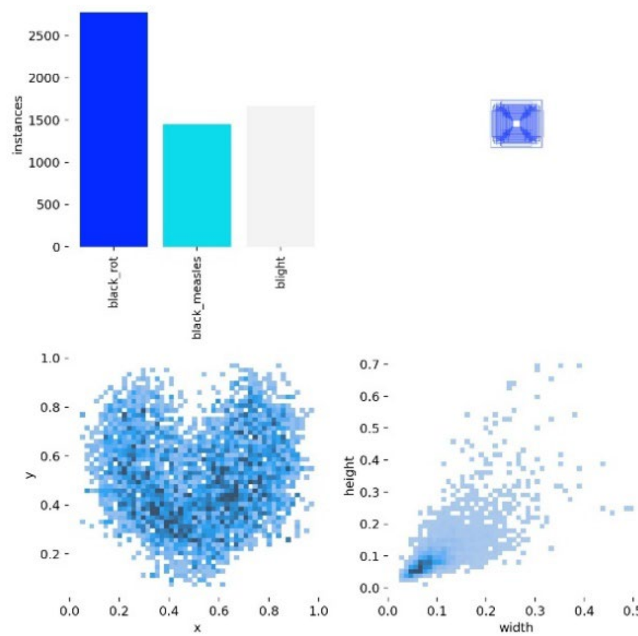


Fig. 3. Target Distribution of the Dataset

## B. Experimental Results and Analysis

Figure 4 shows the F1 score-confidence threshold curves for the YOLOv8 baseline model and the improved model incorporating the SimAM attention mechanism. In the original YOLOv8 model, the average optimal F1 score across all categories was 0.80, corresponding to a confidence threshold of 0.371. The improved YOLOv8-SimAM model elevated the optimal F1 score to 0.81 while lowering the corresponding confidence threshold to 0.329. This change indicates that the SimAM attention mechanism enhances the model's ability to focus on key features, enabling the network to achieve a higher balance between recall and precision at lower confidence thresholds.

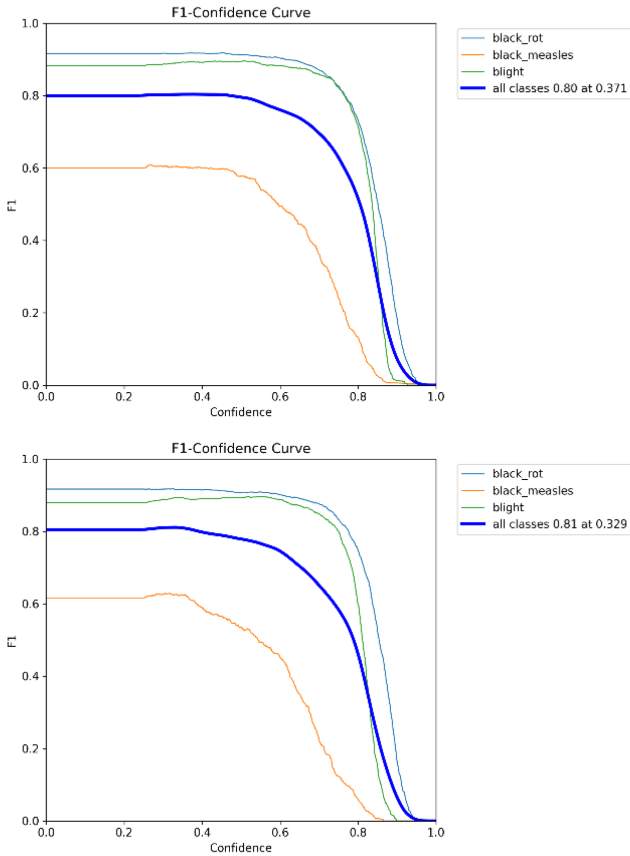


Fig. 4. F1 score-confidence threshold curves for the YOLOv8 baseline model and the improved model

TABLE I. YOLOV8N'S PERFORMANCE ACROSS VARIOUS TESTING METRICS

| objects | P | R | mAP@0.5 | mAP@0.5:0.95 |
|---|---|---|---|---|
| all | **0.797** | 0.81 | 0.816 | 0.528 |
| black_rot | 0.918 | 0.917 | 0.934 | 0.581 |
| black_measles | 0.583 | 0.617 | 0.592 | 0.342 |
| blight | 0.889 | 0.897 | 0.923 | 0.663 |

TABLE II. PERFORMANCE OF YOLOV8N+SIMAM ACROSS VARIOUS DETECTION METRICS

| objects | P | R | mAP@0.5 | mAP@0.5:0.95 |
|---|---|---|---|---|
| all | 0.786 | **0.842** | **0.823** | **0.534** |
| black_rot | 0.916 | 0.918 | 0.935 | 0.581 |
| black_measles | 0.574 | 0.695 | 0.604 | 0.363 |
| blight | 6.867 | 0.913 | 0.929 | 6.657 |

As shown in Table 1 and Table 2, comparing the performance of the YOLOv8n baseline model with the improved model incorporating the SimAM across various detection metrics, the introduction of the SimAM module has significantly optimized model performance. Overall, the improved model increases mAP@0.5 from 0.816 to 0.823, while mAP@0.5:0.95 also improved from 0.528 to 0.534. This indicates that the model enhances both its ability to comprehensively detect targets and its overall detection accuracy while maintaining recognition precision. Category-wise analysis reveals SimAM's most pronounced improvement in the "black measles" category, where recall surged from 0.617 to 0.695, with mAP@0.5 and mAP@0.5:0.95 increasing by 0.012 and 0.021, respectively, effectively mitigating missed detections of this target. For the "blight" category, recall also improved from 0.897 to 0.913, further validating the attention mechanism's role in enhancing the model's ability to distinguish difficult or similar samples.

## V. CONCLUSION

This paper proposes an improved method for intelligent detection of crop pests and diseases by embedding the SimAM attention mechanism across multiple layers within the YOLOv8 network. Experimental results demonstrate that the enhanced model outperforms the original YOLOv8 in metrics including recall, mAP@0.5, and mAP@0.5:0.95, while maintaining its lightweight nature without introducing additional learnable parameters. The YOLOv8-SimAM model demonstrates higher detection accuracy, robustness, and generalization capabilities in crop pest and disease detection. Subsequent research will examine the effectiveness of this approach compared to other attention mechanisms.

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